TX Path Matching:

Match_1_LPF
Match_2_LPF

Si4464 PA Matching Application Note:[AN648]

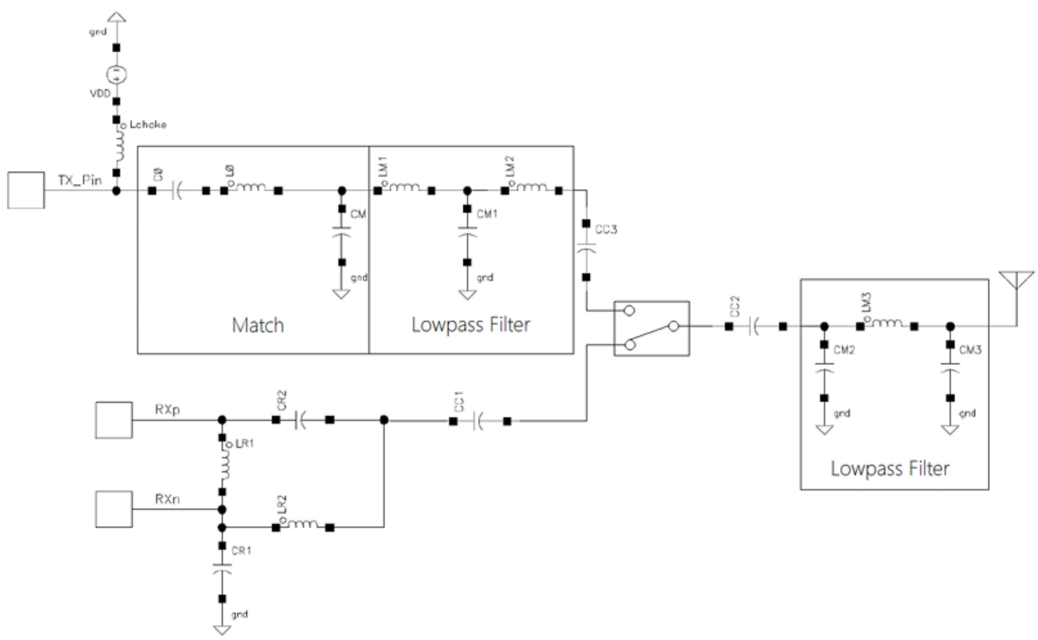

Figure 3. Class-E Matching Topology for Switched TX/RX Board Configuration

Calculations for Input Impedance:

This required PA supply voltage (V_{DD_PA}) is significantly different than the general supply voltage (V_{DD}) for the rest of the RFIC. It is obviously not desirable to maintain two separate, independent sources of supply voltage for the RFIC; however, the PA output supply voltage can be easily created from the main supply voltage by means of an I-R voltage drop across a resistor. This supply voltage drop is accomplished by resistor R_{DC} as shown in Figure 7. As the *theoretical* efficiency of a Class-E switching amplifier is 100%, the average PA drain current I_{DD_PA} may be calculated as:

$$P_{OUT} = \pi\omega_o C_{SHUNT} V_{DD\_PA}^2 = I_{DD\_PA} V_{DD\_PA}$$

Equation 6.

This equation may be solved for I_{DD_PA} to obtain:

$$I_{DD\_PA} = \pi\omega_o C_{SHUNT} V_{DD\_PA}$$

Equation 7.

Given the main supply voltage for the remainder of the chip (V_{DD}), and having previously calculated the required value of PA supply voltage (V_{DD_PA}) and average PA drain current (I_{DD_PA}), the required value for R_{DC} can be calculated as follows:

$$R_{DC} = \frac{V_{DD} - V_{DD\_PA}}{I_{DD\_PA}}$$

Equation 8.

Continuing the design example of 470 MHz for +20 dBm output power, the following calculations are made:

$$I_{DD\_PA} = 2\pi^2 \times 470\text{M} \times 2.5\text{pF} \times 2.076\text{V} = 48.16\text{mA}$$

Equation 9.

Assuming a general chip supply voltage of V_{DD} = 3.3 V, the required value of R_{DC} may be calculated as:

$$R_{DC} = \frac{3.3 - 2.076}{0.04816} = 25.41\Omega$$

Equation 10.

This value of resistance must be placed in series with L_{CHOKE} in order to drop the general chip supply voltage down to the value required to obtain the desired output power.

However, these calculations for the value of R_{DC} assume theoretically-ideal Class-E operation. In practice, the circuit will not behave exactly as predicted by these equations due to non-idealities such as non-zero ON-state resistance, finite OFF-state resistance, non-zero switching times, and loss in the external matching components. As a result, the output power obtained for a given value of V_{DD} supply voltage will almost certainly be slightly less than predicted by theory. Stated another way, it will almost always be necessary to use a smaller value of R_{DC} than predicted by theory in order to obtain the desired value of output power. Some amount of post-calculation "tweaking" of the component values is considered a normal part of the PA matching process.

RX Path Matching:

RX_PATH

Si4464 RX LNA Matching Application Note:[AN643]

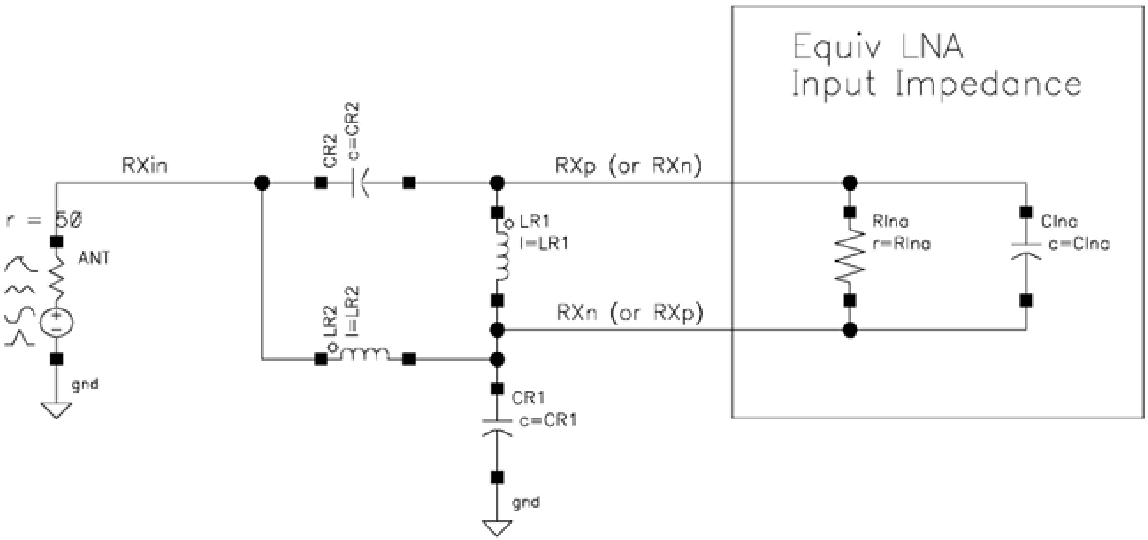

Figure 2. Four-Element Match Network

Table 1. Equivalent R_{LNA}-C_{LNA} from 140-960 MHz

Freq	R_{LNA}	C_{LNA}
140 MHz	545 Ω	1.02 pF
169 MHz	536 Ω	0.98 pF
200 MHz	530 Ω	0.96 pF
250 MHz	520 Ω	0.94 pF
300 MHz	512 Ω	0.95 pF
315 MHz	509 Ω	0.95 pF
350 MHz	499 Ω	0.95 pF
390 MHz	491 Ω	0.96 pF
400 MHz	488 Ω	0.96 pF

Components used with their datasheets: RF Components